PROJECT WORK ACTIVITY

ASSIGNMENT

DESCRIPTION

The company Stark Industries is developing a new solid rocket propulsion system and is characterizing its behavior, mainly looking at the heat load inside the throat.

The Stark-100k is a solid propellant unit giving 100k Newton thrust (sea level). The total impulse is 25 MN.s. It is characterized by an almost constant thrust profile, following the principle of BATES motors (radial + axial burning, granting same initial and final pressure). The pressure is 70 bar and the propellant is a standard non-aluminized AP/HTPB propellant. Nozzle is optimal at sea level.

The company wants to characterize the heat transfer in proximity of the nozzle throat. To do that, the throat does not present the typical insert. Rather, a cooling jacket is used to monitor steady state heat transfer from area ratio 2 (subsonic) to area ratio 2 (supersonic). The jacket is straight and works in co-flow direction, using pressurized water. The transferred heat is derived by monitoring the temperature difference between inlet and outlet of water (consider tap water inlet at 18 Celsius degrees). A thermal barrier is likely to be used, however, its use must be justified with numbers.

The cooling jacket should be reusable. The rest can be shot. The engine is not necessarily optimized for performance: primary target is nozzle hat load quantification.

TASKS

The team of students should perform the following tasks:

- Do a literature search on:
  - thermal barrier coatings, its limits and properties
  - BATES motors, their use and geometries

- Perform the design of the solid propellant system (quasi-steady assumption, no transients)
  - Decide the propellant composition
  - Derive ballistic properties from the attached experimental data report
  - Design geometry of grain and nozzle (Rao or conical, including at least divergence losses)
  - Design thermal protections, according to general regression data available from the literature
  - Characterize the expected real performance (thrust profile, real specific impulse, real total impulse).
  - Give an estimation of uncertainty related MEOP and burning time, assuming constant ideal C* (selection of uncertainty sources, ranges, distributions, ...)
- Design the cooling jacket experimental pack (steady state assumption must be taken)
  - Design the cooling jacket, select nozzle material and thermal barrier (if needed), evaluate expected heat loads, operating pressure of water, operating temperatures ....
  - Design the external supporting apparatus considering that the system should be operated for the whole duration of the fire plus 2 seconds before and after. Consider that water is disposed and, for safety reasons, the system should be independent from external supplies during the operation.

DEADLINES

The team has to deliver a report and Matlab/Octave supporting codes in two different deadlines.

- The report is due on Sunday 19th at 23:59 CEST
- The code is due by May 31st at CEST

CONTENT OF THE REPORT.

PDF file format, max 10 MB, WEBEEP link will be made available.

Technical English language. Max 30 pages long of content. Extra pages apply to appendix, index, reference list, nomenclature and similar. Appendix cannot contain essential material to understand the report and should be used only to complete the reported.

The report is written from technician to technician. We expect to find the literature survey, an essential discussion of theory approach with references, brief requirements discussion, overview of your approach, your engineering choices EXPLAINED and supported, results COMMENTED (and if possible compared against the literature), critical review of results against requirements (if present, comments and justification of deviations), final comments and future developments. No theory explained, unless it is your new personal derivation. If you need extended theory discussion, use appendix and extract in the text only the essential equations.

The report should be readable and understandable (pay attention to quality of images, captions, nomenclature, ...). Also this will be part of the evaluation.

An evaluation grid will be released to guide through as well.

CODING

Each team must deliver a code giving the same results of the report. If you found an error in your results and you correct it, an Erratum PDF document should be attached to the code delivery.

Code and eventual Erratum file must be delivered in one ZIP file (max 10 MB).

Presence of errors will be considered in the correction score. An error not identified in the Erratum is more penalized than a corrected error.

The code must give as explicit result at least:

- plot of the thrust trace of the BATES motor
- uncertainty of rocket ballistic performance, along with convergence monitoring plots on mean and standard deviation.

All other parameters should be computed by the codes. Please, comment the code for better understanding.

The code will be tested on:

- OCTAVE version 11.1.X

- MATLAB version 2025b

APPENDIX

THERMOCOUPLES FOR FLUID MEASUREMENT

You can intall a thermocouple for fluid temperarture monitoring using the principle of the thermal well. Figure 1 is an example of how to do that. You need a T junction, a liquid tight feedthough and a

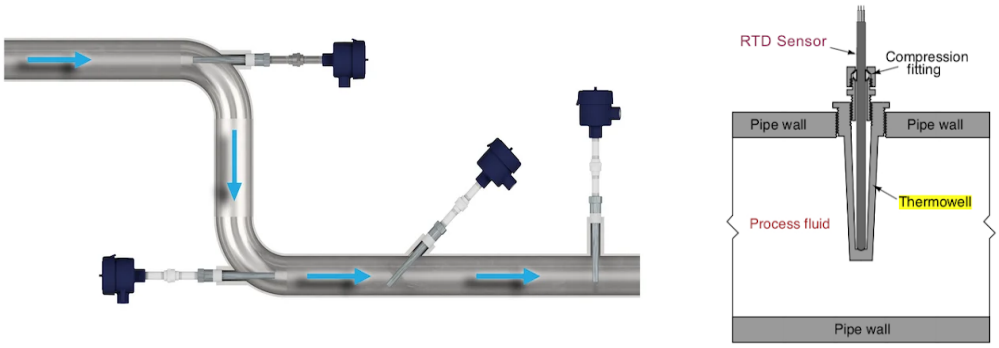


Figure 1: Thermal wellsto measure fluid temperature in a pipe

This information is not to be used for this work. It has only the scope to show you a real implementation of in-line temperature measurements.

Images from:

https://blog.meadobrien.com/2018/09/industrial-thermowells-sometimes-taken.html

https://blog.wika.com/us/knowhow/where-should-i-install-my-thermowell-guide-for-best-practices/

BALLISTIC DATA

The propellant was characterized using a set of small scale BATES motors. The results are here.

Each of these lines is the process of a BATES motor using the same regression algorithm and reporting the effective test pressure and the corresponding burning rate

Pressure, bar	Burning rate, mm/s
10,1	4,0
9,7	3,8
10,3	4,1
30,2	5,6
31,0	6,0
29,8	5,7
50,2	7,0
51,0	7,2
50,3	7,1
70,2	8,4
69,0	8,3
69,8	8,6
91,2	8,8
89,1	9,0
89,0	9,2